

MICHAEL MAINA MWANGI

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TRANSFORMING DATA INTO ACTIONABLE INSIGHTS, DRIVING GROWTH, AND EMPOWERING SMARTER DECISIONS

Analytical and results-driven Data Scientist with a strong foundation in data analytics, analytics engineering, artificial intelligence and business intelligence, delivering innovative solutions that drive business growth and operational excellence. Experienced in creating actionable insights through advanced analytics, data visualization, generative artificial intelligence and machine learning to optimize decision-making and improve customer engagement. Expertise in developing and implementing BI dashboards, machine learning models, and ETL processes to enhance operational efficiency and achieve strategic goals. Proven ability to lead impactful projects, optimize operations processes, and deliver measurable improvements in customer experience and organizational performance. Skilled in fostering data-driven cultures through training programs, empowering teams with enhanced capabilities, and driving a consistent focus on leveraging analytics for business success. Adept at credit scoring, fraud detection, customer segmentation, and digital transformation, with a track record of reducing costs, increasing revenue, and streamlining workflows.

TECHNICAL SKILLS AND TOOLS

- ❖ **Cloud System:** AWS, Google Cloud, Microsoft Azure
- ❖ **Collaboration Tool:** Asana, Jira, Slack, Zoom
- ❖ **Database & SQL:** MariaDB, Microsoft SQL Server, MySQL, Oracle, Postgres, Redis, SQLite
- ❖ **Data Science & Machine Learning:** ETL, Google Colab, Jupyter Notebook, Jupyter Lab, Matplotlib, NLTK, NumPy, Pandas, Power BI, RegEx, Scikit-Learn, SciPy, Seaborn, Snowflake, spaCy, statsmodels
- ❖ **Data Visualization:** Matplotlib, Seaborn, Power BI, Plotly, Looker Studio, Tableau
- ❖ **Design:** Canva
- ❖ **Documentation:** Confluence, Notion
- ❖ **Framework:** Anaconda, Bootstrap, Django, Flask, PyTorch, TensorFlow, Firebase
- ❖ **Operating System:** Windows, MacOS
- ❖ **Package Manager:** PyPI, Yarn, Homebrew
- ❖ **Programming Language:** Bash, CSS, HTML, JavaScript, JSON, Markdown, Python, R, YAML, TailwindCSS, Apache Spark (Pyspark), Power BI DAX
- ❖ **Productivity & Workflow:** Bitbucket, MS Office
- ❖ **Version Control:** Git
- ❖ **Web Scraping & APIs:** Requests, Beautiful Soup, FastAPI

CORE COMPETENCIES

Business Intelligence & Data Reporting | Data Science & Engineering | Machine Learning Model Development | Financial & Operational Analytics | Customer Segmentation & Fraud Detection | BI Dashboards & Data Visualization | Team Training & Development | Strategic Planning & Decision-Making | Process Optimization & Efficiency | Credit Management & Loan Recovery Systems | AI-Powered Solutions | Design

CAREER SUMMARY

I&M Group PLC | November 2025 – Present

Senior Data Scientist – Enterprise Data Office

Key Highlights:

- Designed and deployed an early warning system for the group's corporate banking customers specific to the overdraft lending portfolio with a catch rate of 83% for customers likely to become delinquent.
- Developed Generative AI applications for different departments in the group to improve situations such as on OCR application designed to speed up data gathering from conveyancing documents for the legal department by up to 7 times faster than manual data capture.
- Developed a fraud detection machine learning model deployed to rate fraud probability for corporate customer transactions on the bank's payment platforms.
- Boosted team's organizational proficiency by 60% by developing and designing the team's data catalog and metadata manager for developed use cases.

Equity Group Holdings | April 2024 – November 2025

Assistant Manager – Data Science and Engineering

Key Highlights:

- Developed Generative AI applications for different departments in the group to improve situations such as quicker onboarding of new staff through the Human Resources Department. Other applications have been vital to improve staff productivity e.g. faster access to documentation details and access to other knowledgebases.
- Developed an automation tool that batch proofs salary reconciliations for the digital operations department. Led to 50% increase in staff assignment optimization and 80% reduction in proofing times.
- Processed and analyzed 500M+ card transactions using Python (Pandas, NumPy), SQL & Power BI, generating insights that optimized card spending by 50%, increased new card user acquisition by 20%, reduced transaction failure rates from 5% to 0.45% and enhanced business growth through collaborative marketing partnerships with top merchants.
- Boosted team proficiency by 30% and reduced the onboarding time for new hires by enhancing team capability through designing and delivering specialized data science training.
- Designed a credit monitoring dashboard using Power BI and SQL, integrating real-time data to proactively reduce default rates by 50% through enhanced risk management strategies. → Consolidated data for actionable insights that reduced branch service time by 50% and enhanced customer satisfaction by developing a 360-degree customer insights dashboard using Power BI and SQL.

Equity Group Holdings | August 2022 – April 2024

Data Scientist – Data Science and Engineering

Key Highlights:

- Enhanced excellence, ensuring a 10% deviation from the financial target while driving cost-efficiency and supporting strategic financial planning by engineering machine learning models that optimized banking costs.
- Empowered data-driven decision-making to expand revenue streams and strengthen customer engagement through creating BI reports and dashboards that boosted digital lending by 20%+.
- Streamlined service delivery and improved customer satisfaction by enabling quicker response times and actionable insights through explainable BI dashboards to address customer issues.
- Improved operational efficiency by conducting impactful BI training for new and existing staff, fostering a data-driven culture.
- Drove business growth and aligned with long-term organizational objectives by contributing to strategic goals while expanding the customer base and loan portfolio.
- Heightened operational efficiency and enhanced customer experience by designing dashboards to analyze channel usage, enabling the identification and resolution of key business challenges.

Equity Bank Kenya | August 2019 – August 2022

Junior Analytics Engineer – Advanced Analytics

Key Highlights:

- Improved fraud prevention accuracy and reduced operational losses by engineered fraud detection models with a false positive rate under 20% and a true positive rate exceeding 80%.
- Enabled more targeted risk mitigation strategies, performing customer data analytics to enhance segmentation and strengthen fraud tracking in payment systems.
- Recorded 20% annual reduction in branch visits through increased digital service adoption and enhanced operational efficiency through crafted dashboards for Operations.
- Enriched product offerings and boosted customer retention following impactful executive presentations by developing a Customer Product Basket App.
- Kept faster turnaround times and improved productivity, streamlining ETL processes, dramatically increasing the efficiency of departmental ad hoc data tasks.

EARLY CAREER EXPERIENCE

Actuarial Analyst | Strathmore University – Risk Management Center | 2017

Actuarial Analyst | Kenbright Actuarial and Financial Services | 2017

EDUCATION

Certificate in Data Science | Strathmore University | 2019

Bachelor of Business Science in Actuarial Science | Strathmore University | 2018

CERTIFICATIONS

Generative AI Leader | Google Cloud | October 2025

Relational Database Developer | freeCodeCamp | April 2025

Microsoft Certified: Power BI Data Analyst Associate | Microsoft | December 2022

Professional Data Analyst | Datacamp | July 2022

AI Ethics Certification | Udemy | June 2022

Microsoft Certified: Power Platform Fundamentals | Microsoft | December 2021

PROJECTS

AI Human Resource Business Person | Apr 2025

The Cardinals of Africa | Feb 2024 - Apr 2024

Time Series Analysis of the USD/KES FX Rate | Mar 2023

Cash Management Optimization | Apr 2023 - Dec 2023

Spotify Music Data Analysis | Jun 2022 – Oct 2022

Hotel Booking Demand | Oct 2021 - Nov 2021

Customer Product Basket App | Sep 2021 - Jun 2023